```
import pandas as pd
```

```
df=pd.read_csv('/content/Heart_Disease_Prediction.csv')
df
```

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression	Slope of ST	Number of vessels fluro	Thallium	Heart Disease
0	70	1	4	130	322	0	2	109	0	2.4	2	3	3	Presence
1	67	0	3	115	564	0	2	160	0	1.6	2	0	7	Absence
2	57	1	2	124	261	0	0	141	0	0.3	1	0	7	Presence
3	64	1	4	128	263	0	0	105	1	0.2	2	1	7	Absence
4	74	0	2	120	269	0	2	121	1	0.2	1	1	3	Absence
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
265	52	1	3	172	199	1	0	162	0	0.5	1	0	7	Absence
266	44	1	2	120	263	0	0	173	0	0.0	1	0	7	Absence
267	56	0	2	140	294	0	2	153	0	1.3	2	0	3	Absence
268	57	1	4	140	192	0	0	148	0	0.4	2	0	6	Absence
269	67	1	4	160	286	0	2	108	1	1.5	2	3	3	Presence

270 rows × 14 columns

```
df.head()
```

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression	Slope of ST	Number of vessels fluro	Thallium	Heart Disease
0	70	1	4	130	322	0	2	109	0	2.4	2	3	3	Presence
1	67	0	3	115	564	0	2	160	0	1.6	2	0	7	Absence
2	57	1	2	124	261	0	0	141	0	0.3	1	0	7	Presence
3	64	1	4	128	263	0	0	105	1	0.2	2	1	7	Absence
4	74	0	2	120	269	0	2	121	1	0.2	1	1	3	Absence

```
df.tail()
```

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression	Slope of ST	Number of vessels fluro	Thallium	Heart Disease
265	52	1	3	172	199	1	0	162	0	0.5	1	0	7	Absence
266	44	1	2	120	263	0	0	173	0	0.0	1	0	7	Absence
267	56	0	2	140	294	0	2	153	0	1.3	2	0	3	Absence
268	57	1	4	140	192	0	0	148	0	0.4	2	0	6	Absence
269	67	1	4	160	286	0	2	108	1	1.5	2	3	3	Presence

```
df.shape
```

```
(270, 14)
```

```
df.columns
```

```
Index(['Age', 'Sex', 'Chest pain type', 'BP', 'Cholesterol', 'FBS over 120',
       'EKG results', 'Max HR', 'Exercise angina', 'ST depression',
       'Slope of ST', 'Number of vessels fluro', 'Thallium', 'Heart Disease'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 270 entries, 0 to 269
Data columns (total 14 columns):
 #   Column                   Non-Null Count  Dtype  
---  ------                   --------------  -----  
 0   Age                      270 non-null    int64  
 1   Sex                      270 non-null    int64  
 2   Chest pain type          270 non-null    int64  
 3   BP                       270 non-null    int64  
 4   Cholesterol              270 non-null    int64  
 5   FBS over 120             270 non-null    int64  
 6   EKG results              270 non-null    int64  
 7   Max HR                   270 non-null    int64  
 8   Exercise angina          270 non-null    int64  
 9   ST depression            270 non-null    float64
 10  Slope of ST              270 non-null    int64  
 11  Number of vessels fluro  270 non-null    int64  
 12  Thallium                 270 non-null    int64  
 13  Heart Disease            270 non-null    object 
dtypes: float64(1), int64(12), object(1)
memory usage: 29.7+ KB
```

```
df.dtypes
```

	0
Age	int64
Sex	int64
Chest pain type	int64
BP	int64
Cholesterol	int64
FBS over 120	int64
EKG results	int64
Max HR	int64
Exercise angina	int64
ST depression	float64
Slope of ST	int64
Number of vessels fluro	int64
Thallium	int64
Heart Disease	object

dtype: object

```
numerical_features = df.select_dtypes(include=['int64', 'float64'])
numerical_features.columns
```

```
Index(['Age', 'Sex', 'Chest pain type', 'BP', 'Cholesterol', 'FBS over 120',
       'EKG results', 'Max HR', 'Exercise angina', 'ST depression',
       'Slope of ST', 'Number of vessels fluro', 'Thallium'],
      dtype='object')
```

```
categorical_features = df.select_dtypes(include=['object'])
categorical_features.columns
```

```
Index(['Heart Disease'], dtype='object')
```

```
cholesterol_array=df['Cholesterol'].to_numpy()
cholesterol_array
```

```
array([322, 564, 261, 263, 269, 177, 256, 239, 293, 407, 234, 226, 235,
       234, 303, 149, 311, 203, 211, 199, 229, 245, 303, 204, 288, 275,
       243, 295, 230, 265, 229, 228, 215, 326, 200, 256, 207, 273, 180,
       222, 223, 209, 233, 197, 218, 211, 149, 197, 246, 225, 315, 205,
       417, 195, 234, 198, 166, 178, 249, 281, 126, 305, 226, 240, 233,
       276, 261, 319, 242, 243, 260, 354, 245, 197, 223, 309, 208, 199,
       209, 236, 218, 198, 270, 214, 201, 244, 208, 270, 306, 243, 221,
       330, 266, 206, 212, 275, 302, 234, 313, 244, 141, 237, 269, 289,
       254, 274, 222, 258, 177, 160, 327, 235, 305, 304, 295, 271, 249,
       288, 226, 283, 188, 286, 274, 360, 273, 201, 267, 196, 201, 230,
       269, 212, 226, 246, 232, 177, 277, 249, 210, 207, 212, 271, 233,
       213, 283, 282, 230, 167, 224, 268, 250, 219, 267, 303, 256, 204,
       217, 308, 193, 228, 231, 244, 262, 259, 211, 325, 254, 197, 236,
       282, 234, 254, 299, 211, 182, 294, 298, 231, 254, 196, 240, 409,
```

```
       172, 265, 246, 315, 184, 233, 394, 269, 239, 174, 309, 282, 255,
       250, 248, 214, 239, 304, 277, 300, 258, 299, 289, 298, 318, 240,
       309, 250, 288, 245, 213, 216, 204, 204, 252, 227, 258, 220, 239,
       254, 168, 330, 183, 203, 263, 341, 283, 186, 307, 219, 260, 255,
       231, 164, 234, 177, 257, 325, 274, 321, 264, 268, 308, 253, 248,
       269, 185, 282, 188, 219, 290, 175, 212, 302, 243, 353, 335, 247,
       340, 206, 284, 266, 229, 199, 263, 294, 192, 286])
```

```
df.isnull().sum()
```

	0
Age	0
Sex	0
Chest pain type	0
BP	0
Cholesterol	0
FBS over 120	0
EKG results	0
Max HR	0
Exercise angina	0
ST depression	0
Slope of ST	0
Number of vessels fluro	0
Thallium	0
Heart Disease	0

dtype: int64

checking null values

```
df.isnull().values.any()
```

```
np.False_
```

Row wise checking null values

```
df[df.isnull().any(axis=1)]
```

Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression	Slope of ST	Number of vessels fluro	Thallium	Heart Disease

```
df.describe()
```

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression	Slope of ST	Number of vessels fluro	Thallium
count	270.000000	270.000000	270.000000	270.000000	270.000000	270.000000	270.000000	270.000000	270.000000	270.00000	270.000000	270.000000	270.000000
mean	54.433333	0.677778	3.174074	131.344444	249.659259	0.148148	1.022222	149.677778	0.329630	1.05000	1.585185	0.670370	4.696296
std	9.109067	0.468195	0.950090	17.861608	51.686237	0.355906	0.997891	23.165717	0.470952	1.14521	0.614390	0.943896	1.940659
min	29.000000	0.000000	1.000000	94.000000	126.000000	0.000000	0.000000	71.000000	0.000000	0.00000	1.000000	0.000000	3.000000
25%	48.000000	0.000000	3.000000	120.000000	213.000000	0.000000	0.000000	133.000000	0.000000	0.00000	1.000000	0.000000	3.000000
50%	55.000000	1.000000	3.000000	130.000000	245.000000	0.000000	2.000000	153.500000	0.000000	0.80000	2.000000	0.000000	3.000000
75%	61.000000	1.000000	4.000000	140.000000	280.000000	0.000000	2.000000	166.000000	1.000000	1.60000	2.000000	1.000000	7.000000
max	77.000000	1.000000	4.000000	200.000000	564.000000	1.000000	2.000000	202.000000	1.000000	6.20000	3.000000	3.000000	7.000000

```
import numpy as np
```

```
age_array=df['Age'].to_numpy()
age_array
```

```
array([70, 67, 57, 64, 74, 65, 56, 59, 60, 63, 59, 53, 44, 61, 57, 71, 46,
       53, 64, 40, 67, 48, 43, 47, 54, 48, 46, 51, 58, 71, 57, 66, 37, 59,
       50, 48, 61, 59, 42, 48, 40, 62, 44, 46, 59, 58, 49, 44, 66, 65, 42,
       52, 65, 63, 45, 41, 61, 60, 59, 62, 57, 51, 44, 60, 63, 57, 51, 58,
       44, 47, 61, 57, 70, 76, 67, 45, 45, 39, 42, 56, 58, 35, 58, 41, 57,
       42, 62, 59, 41, 50, 59, 61, 54, 54, 52, 47, 66, 58, 64, 50, 44, 67,
       49, 57, 63, 48, 51, 60, 59, 45, 55, 41, 60, 54, 42, 49, 46, 56, 66,
       56, 49, 54, 57, 65, 54, 54, 62, 52, 52, 60, 63, 66, 42, 64, 54, 46,
       67, 56, 34, 57, 64, 59, 50, 51, 54, 53, 52, 40, 58, 41, 41, 50, 54,
       64, 51, 46, 55, 45, 56, 66, 38, 62, 55, 58, 43, 64, 50, 53, 45, 65,
       69, 69, 67, 68, 34, 62, 51, 46, 67, 50, 42, 56, 41, 42, 53, 43, 56,
       52, 62, 70, 54, 70, 54, 35, 48, 55, 58, 54, 69, 77, 68, 58, 60, 51,
       55, 52, 60, 58, 64, 37, 59, 51, 43, 58, 29, 41, 63, 51, 54, 44, 54,
       65, 57, 63, 35, 41, 62, 43, 58, 52, 61, 39, 45, 52, 62, 62, 53, 43,
       47, 52, 68, 39, 53, 62, 51, 60, 65, 65, 60, 60, 54, 44, 44, 51, 59,
       71, 61, 55, 64, 43, 58, 60, 58, 49, 48, 52, 44, 56, 57, 67])
```

```
age_array=df['Age'].to_numpy().reshape(-1,1)
age_array
```

```
array([[70],
       [67],
       [57],
       [64],
       [74],
       [65],
       [56],
       [59],
       [60],
       [63],
       [59],
       [53],
       [44],
       [61],
       [57],
```

```
        [71],
        [46],
        [53],
        [64],
        [40],
        [67],
        [48],
        [43],
        [47],
        [54],
        [48],
        [46],
        [51],
        [58],
        [71],
        [57],
        [66],
        [37],
        [59],
        [50],
        [48],
        [61],
        [59],
        [42],
        [48],
        [40],
        [62],
        [44],
        [46],
        [59],
        [58],
        [49],
        [44],
        [66],
        [65],
        [42],
        [52],
        [65],
        [63],
        [45],
        [41],
        [61],
        [60],
```

```
mean_age=np.mean(age_array)
median_age=np.median(age_array)
std_age=np.std(age_array)
print(mean_age,',',median_age,',',std_age)
```

```
54.43333333333333 , 55.0 , 9.092182234083177
```

```
cholesterols=df['Cholesterol'].to_numpy()
cholesterols
maximum_values=np.max(cholesterols)
minimum_values=np.min(cholesterols)
print(maximum_values,',',minimum_values)
```

```
564 , 126
```

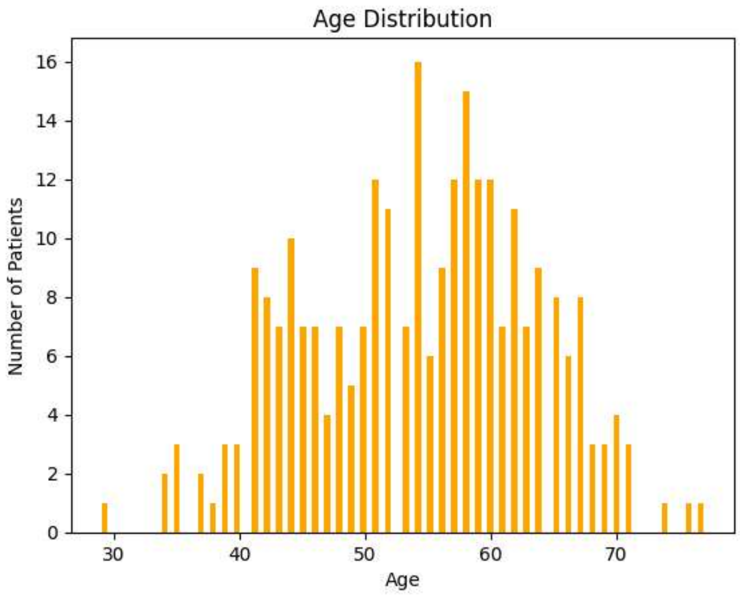
```
count_hd=df['Heart Disease'].value_counts()
count_hd
```

Heart Disease	count
Absence	150
Presence	120

dtype: int64

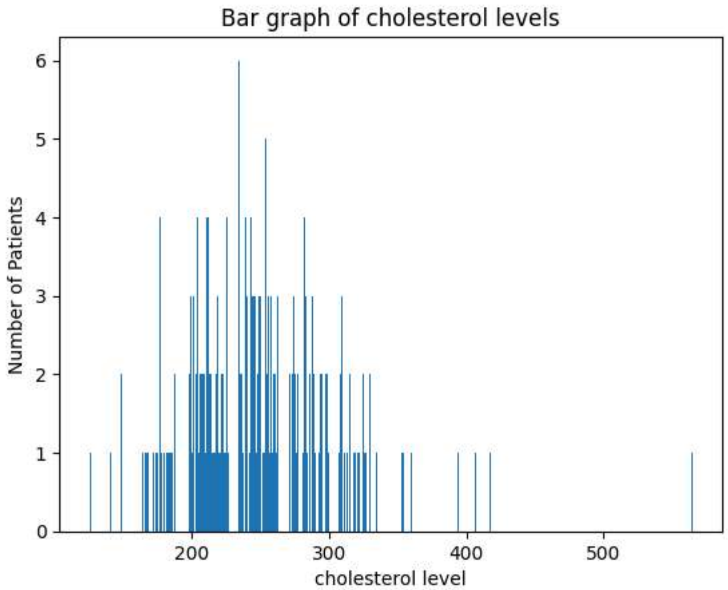
```
import matplotlib.pyplot as plt
```

```
plt.hist(age_array,color='Orange',bins=100)
plt.xlabel('Age')
plt.ylabel('Number of Patients')
plt.title('Age Distribution')
plt.show()
```

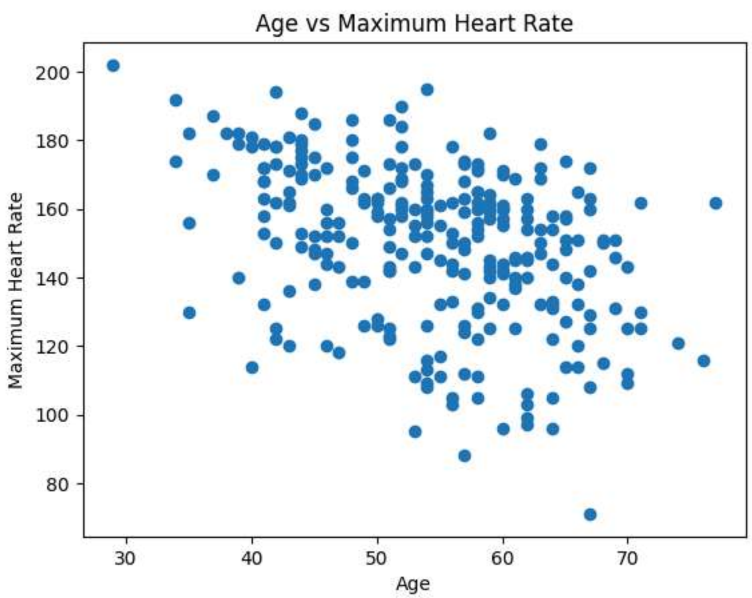



```
values,count=np.unique(cholesterol_array,return_counts=True)
plt.bar(values,count)
plt.xlabel('cholesterol level')
plt.ylabel('Number of Patients')
plt.title('Bar graph of cholesterol levels')
```

```
Text(0.5, 1.0, 'Bar graph of cholesterol levels')
```




```
plt.scatter(df['Age'],df['Max HR'])
plt.xlabel('Age')
plt.ylabel('Maximum Heart Rate')
plt.title('Age vs Maximum Heart Rate')
plt.show()
```



```
df_new=df.drop(['Heart Disease'],axis=1)
df_new
```

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression	Slope of ST	Number of vessels fluro	Thallium
0	70	1	4	130	322	0	2	109	0	2.4	2	3	3
1	67	0	3	115	564	0	2	160	0	1.6	2	0	7
2	57	1	2	124	261	0	0	141	0	0.3	1	0	7
3	64	1	4	128	263	0	0	105	1	0.2	2	1	7
4	74	0	2	120	269	0	2	121	1	0.2	1	1	3
...	...	...	...	...	...	...	...	...	...	...	...	...	...
265	52	1	3	172	199	1	0	162	0	0.5	1	0	7
266	44	1	2	120	263	0	0	173	0	0.0	1	0	7
267	56	0	2	140	294	0	2	153	0	1.3	2	0	3
268	57	1	4	140	192	0	0	148	0	0.4	2	0	6
269	67	1	4	160	286	0	2	108	1	1.5	2	3	3

270 rows × 13 columns

```
correlation_matrix=df_new.corr()
correlation_matrix
```

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression	Slope of ST	Number of vessels fluro	Thallium
Age	1.000000	-0.094401	0.096920	0.273053	0.220056	0.123458	0.128171	-0.402215	0.098297	0.194234	0.159774	0.356081	0.106100
Sex	-0.094401	1.000000	0.034636	-0.062693	-0.201647	0.042140	0.039253	-0.076101	0.180022	0.097412	0.050545	0.086830	0.391046
Chest pain type	0.096920	0.034636	1.000000	-0.043196	0.090465	-0.098537	0.074325	-0.317682	0.353160	0.167244	0.136900	0.225890	0.262659
BP	0.273053	-0.062693	-0.043196	1.000000	0.173019	0.155681	0.116157	-0.039136	0.082793	0.222800	0.142472	0.085697	0.132045
Cholesterol	0.220056	-0.201647	0.090465	0.173019	1.000000	0.025186	0.167652	-0.018739	0.078243	0.027709	-0.005755	0.126541	0.028836
FBS over 120	0.123458	0.042140	-0.098537	0.155681	0.025186	1.000000	0.053499	0.022494	-0.004107	-0.025538	0.044076	0.123774	0.049237
EKG results	0.128171	0.039253	0.074325	0.116157	0.167652	0.053499	1.000000	-0.074628	0.095098	0.120034	0.160614	0.114368	0.007337
Max HR	-0.402215	-0.076101	-0.317682	-0.039136	-0.018739	0.022494	-0.074628	1.000000	-0.380719	-0.349045	-0.386847	-0.265333	-0.253397
Exercise angina	0.098297	0.180022	0.353160	0.082793	0.078243	-0.004107	0.095098	-0.380719	1.000000	0.274672	0.255908	0.153347	0.321449
ST depression	0.194234	0.097412	0.167244	0.222800	0.027709	-0.025538	0.120034	-0.349045	0.274672	1.000000	0.609712	0.255005	0.324333
Slope of ST	0.159774	0.050545	0.136900	0.142472	-0.005755	0.044076	0.160614	-0.386847	0.255908	0.609712	1.000000	0.109498	0.283678
Number of vessels fluro	0.356081	0.086830	0.225890	0.085697	0.126541	0.123774	0.114368	-0.265333	0.153347	0.255005	0.109498	1.000000	0.255648
Thallium	0.106100	0.391046	0.262659	0.132045	0.028836	0.049237	0.007337	-0.253397	0.321449	0.324333	0.283678	0.255648	1.000000

```
correlation_matrix.round(2)
```

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression	Slope of ST	Number of vessels fluro	Thallium
Age	1.00	-0.09	0.10	0.27	0.22	0.12	0.13	-0.40	0.10	0.19	0.16	0.36	0.11
Sex	-0.09	1.00	0.03	-0.06	-0.20	0.04	0.04	-0.08	0.18	0.10	0.05	0.09	0.39
Chest pain type	0.10	0.03	1.00	-0.04	0.09	-0.10	0.07	-0.32	0.35	0.17	0.14	0.23	0.26
BP	0.27	-0.06	-0.04	1.00	0.17	0.16	0.12	-0.04	0.08	0.22	0.14	0.09	0.13
Cholesterol	0.22	-0.20	0.09	0.17	1.00	0.03	0.17	-0.02	0.08	0.03	-0.01	0.13	0.03
FBS over 120	0.12	0.04	-0.10	0.16	0.03	1.00	0.05	0.02	-0.00	-0.03	0.04	0.12	0.05
EKG results	0.13	0.04	0.07	0.12	0.17	0.05	1.00	-0.07	0.10	0.12	0.16	0.11	0.01
Max HR	-0.40	-0.08	-0.32	-0.04	-0.02	0.02	-0.07	1.00	-0.38	-0.35	-0.39	-0.27	-0.25
Exercise angina	0.10	0.18	0.35	0.08	0.08	-0.00	0.10	-0.38	1.00	0.27	0.26	0.15	0.32
ST depression	0.19	0.10	0.17	0.22	0.03	-0.03	0.12	-0.35	0.27	1.00	0.61	0.26	0.32
Slope of ST	0.16	0.05	0.14	0.14	-0.01	0.04	0.16	-0.39	0.26	0.61	1.00	0.11	0.28
Number of vessels fluro	0.36	0.09	0.23	0.09	0.13	0.12	0.11	-0.27	0.15	0.26	0.11	1.00	0.26
Thallium	0.11	0.39	0.26	0.13	0.03	0.05	0.01	-0.25	0.32	0.32	0.28	0.26	1.00

```
high_corr=correlation_matrix.abs()>0.8
high_corr
```

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression	Slope of ST	Number of vessels fluro	Thallium
Age	True	False	False	False	False	False	False	False	False	False	False	False	False
Sex	False	True	False	False	False	False	False	False	False	False	False	False	False
Chest pain type	False	False	True	False	False	False	False	False	False	False	False	False	False
BP	False	False	False	True	False	False	False	False	False	False	False	False	False
Cholesterol	False	False	False	False	True	False	False	False	False	False	False	False	False
FBS over 120	False	False	False	False	False	True	False	False	False	False	False	False	False